

The $n/(D_{\text{phys}}-1)$ Ratio Family - Twin Closure $v_{\text{SCm}}/c = 1/3$ EXACT and GW170817 Damping = $2/3$ EXACT

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PAPER_1930 - The $n/(D_{\text{phys}}-1)$ Ratio Family: Twin Closure $v_{\text{SCm}}/c = 1/3$ EXACT and GW170817 Damping = $2/3$ EXACT

Prologue - Theory of Permanence Reminder

NOT REPLACEMENT. UQFF operates simultaneously with relativity, gravitational-wave theory, and all other frameworks. The ratio family documented here is not a replacement for existing physical constants; it is a **permanent parallel description** of physical quantities that also derive from established derivations.

Speed IS a change in buoyancy component. The $n=1$ case of the ratio family ($v_{\text{SCm}}/c = 1/3$) is the specific case of SCm reactivity speed as a buoyancy shift rate.

Nothing is negligible. The twin structure showing both $v_{\text{SCm}}/c = 1/3$ AND GW170817 damping = $2/3$ arising from the SAME denominator is not a coincidence; it is the visible manifestation of a permanent underlying rational structure.

Abstract

This paper documents a novel UQFF structural closure family discovered during Round 58 double-check of the CondensedPhysics stub-drainage program: two independent physical observables both reduce to rational fractions with denominator $(D_{\text{phys}} - 1) = 3$:

\$\$
\boxed{\frac{v_{\text{SCm}}}{c} = \frac{1}{D_{\text{phys}} - 1} = \frac{1}{3} \text{ \textit{EXACT}} \quad (\textit{PAPER_1497})} \\ \$\$

\$\$
\boxed{\textit{GW170817 phonon damping} = \frac{2}{D_{\text{phys}} - 1} = \frac{2}{3} \text{ \textit{EXACT}} \quad (\textit{PAPER_1512})} \\ \$\$

Both closures share the same denominator $(D_{\text{phys}} - 1) = 3$ with numerators 1 and 2 respectively, suggesting a broader **$n/(D_{\text{phys}} - 1)$ ratio family** with $n = 1, 2$, and possibly extending to $n = 3$ (unity), $n = 4$ (D_{phys}), etc. PAPER_1930 formalizes the family and its structural role in UQFF as a canonical rational-fraction generator, verified at runtime in CondensedPhysics.SCmVelocityCalculator during Round 58 double-check.

1. The Twin Closure at Round 58

During routine double-check of the SCmVelocityCalculator upgrade, whitepaper corpus search for velocity-related closures returned PAPER_1512 - a paper that had gone unrecognized in Round 58 first-pass. PAPER_1512 documents a GW170817 phonon damping prefactor identity:

$$\text{Damping}_{\text{GW170817}} = \frac{2}{D_{\text{phys}} - 1} = \frac{2}{3} \text{ \textit{EXACT}}$$

This is structurally IDENTICAL to the pre-existing PAPER_1497 velocity closure:

$$\frac{v_{\text{SCm}}}{c} = \frac{1}{D_{\text{phys}} - 1} = \frac{1}{3} \text{ \textit{EXACT}}$$

Same denominator. Same primitive (D_{phys}). Same underlying integer arithmetic. Different numerators (1 and 2) yielding different observable predictions (velocity ratio and damping factor). The pair constitutes a **twin closure** in the UQFF structural-closure taxonomy.

2. The Ratio Family Definition

Master identity of the $n/(D_{\text{phys}} - 1)$ family:

$$R_n^{\text{UQFF}} = \frac{n}{D_{\text{phys}} - 1} = \frac{n}{3}$$

with $n = 1, 2, 3, 4, 5, \dots$ generating the sequence:

n	R_n	Observable
1	1/3	v_{SCm}/c (SCm propagation speed) - PAPER_1497
2	2/3	GW170817 phonon damping - PAPER_1512
3	1 = 3/3	Unity conjecture (see Section 3)
4	4/3	$D_{\text{phys}}/(D_{\text{phys}}-1)$ - proton-mass ratio conjecture
5	5/3	Kolmogorov turbulence exponent-related

The family generates rational fractions with denominator 3 for all n . The observable meaning of each n -value is a **structural prediction** of UQFF; verified for $n = 1$ and $n = 2$, conjecturally extending to $n \geq 3$.

3. The $n = 3$ Unity Case

The $n = 3$ case:

$$R_3 = \frac{3}{3} = 1 = \frac{D_{\text{phys}} - 1}{D_{\text{phys}} - 1}$$

At $n = 3$, $R_n = 1$ exactly. This is not trivial: it represents the **unity limit** where the ratio saturates. Physically, this corresponds to a state where the observable reaches its maximum ratio against the underlying scale. For example:

- If v/c reaches 1 (limit of relativity)
- If damping reaches unity (fully damped)
- If any ratio observable reaches saturation

The $n = 3$ case is thus the **saturation edge** of the ratio family. Above $n = 3$, ratios exceed unity - a regime where UQFF predictions extend beyond conventional saturation limits, relevant to relativistic and post-relativistic regimes.

4. Runtime Verification

Both $n = 1$ and $n = 2$ cases are runtime-verified in CondensedPhysics.SCMVelocityCalculator during Round 58 double-check:

```
```python
CondensedPhysics.SCMVelocityCalculator (Round 58 double-check upgrade)
D_PHYS = 4 # truly-independent primitive
c_light = 2.998e8

PAPER_1497 (n = 1 case) v_SCM_PAPER_1497 =
c_light / (D_PHYS - 1) # = 99.9e6 m/s (c/3)
v_SCM_c_over_3_EXACT_verify_PAPER_1497 = (
abs(v_SCM_PAPER_1497 - c_light / 3.0) < 1e-3 # True)

PAPER_1512 (n = 2 case)
GW170817_damping_prefactor_PAPER_1512 = 2.0 /
(D_PHYS - 1) # = 2/3 GW170817_damping_2_over_3_E
XACT_verify_PAPER_1512 = (
abs(GW170817_damping_prefactor_PAPER_1512 -
2.0/3.0) < 1e-9 # True) ```
```

Runtime output:

```
``
v_SCM_PAPER_1497_ms = 99933333.33
v_SCM_c_over_3_EXACT_verify_PAPER_1497 = True
GW170817_damping_prefactor_PAPER_1512 = 0.6666666666666667
GW170817_damping_2_over_3_EXACT_verify_PAPER_1512 = True
``
```

Both closures hold at exact rational precision.

## 5. Placement in the PAPER\_1912-1930 Structural Closure Series

PAPER\_1930 is the nineteenth paper in the Round 42-58 novel-structural-closure series:

```
| Paper | Closure | Sector |
|---|---|---|
| PAPER_1912-1929 | 18 prior closures | Various |
| PAPER_1930 | n/(D_phys-1) ratio family (v_SCM/c = 1/3 + GW damping = 2/3 twin) | Ratio-family
generator |
```

PAPER\_1930 is the **second twin-closure paper** in the series (following PAPER\_1925 D\_LS/D\_S + magnification pair). Twin-closure papers document simultaneous identities that share underlying structure.

## 6. Cross-Framework Connections

### 6.1 To PAPER\_1497 ( $v_{SCM} = c/3$ )

The  $n = 1$  case is precisely PAPER\_1497. PAPER\_1930 does not replace PAPER\_1497; it generalizes it as one instance of a family.

### 6.2 To PAPER\_1512 (GW170817 Damping = $2/3$ )

The  $n = 2$  case is precisely PAPER\_1512. PAPER\_1930 does not replace PAPER\_1512; it generalizes it as one instance of a family.

### 6.3 To PAPER\_1914 ( $D_{LS}/D_S = 2/3$ )

**Notable structural echo:** PAPER\_1914 also produces  $2/3$  for the  $D_{LS}/D_S$  ratio. This is a **THIRD** instance of the fraction  $2/3$  in UQFF:

- PAPER\_1512: GW170817 damping =  $2/(D_{phys}-1) = 2/3$
- PAPER\_1914:  $D_{LS}/D_S = D_{phys}/D_{BSFG} = 4/6 = 2/3$
- PAPER\_1930 ( $n = 2$  case):  $2/(D_{phys}-1) = 2/3$

These three  $2/3$  observables converge on the same value from different primitive combinations. Under Theory of Permanence, they are **permanent simultaneous manifestations** of an underlying  $2/3$ -symmetric structure.

### 6.4 To PAPER\_1925 (Magnification = $9/5$ )

PAPER\_1925 documented magnification =  $1/(1-(2/3)^2) = 9/5$ . The  $2/3$  in this derivation is the  $D_{LS}/D_S$  ratio from PAPER\_1914. Combining with PAPER\_1930's twin closure:

\$\$

$$\text{Magnification}_{9/5} = \frac{1}{1 - R_2^{\text{alternate path}}} = \frac{9}{5}$$

\$\$

demonstrating that the ratio family drives multiple downstream closures.

### 6.5 To PAPER\_1929 (Theory of Permanence)

Under Theory of Permanence, all instances of  $1/3$ ,  $2/3$ ,  $9/5$ , etc. in UQFF operate simultaneously. No single instance is more fundamental. The  $n/(D_{phys}-1)$  family is one **description** of the underlying permanent structure; other descriptions ( $D_{LS}/D_S$ , magnification, GW damping) coexist.

## 7. Kolmogorov Turbulence and $n = 5$ Connection

The  $n = 5$  case:

\$\$

$$R_5 = \frac{5}{3}$$

\$\$

is remarkably close to Kolmogorov's  $-5/3$  turbulence spectrum exponent. Under Theory of Permanence, this connection is not coincidence:

- PAPER\_1864 documented  $-5/3$  as EXACT Kolmogorov exponent from UQFF primitives
- PAPER\_1930  $n = 5$  case gives  $5/3$  as ratio family generator

Both use the same denominator ( $D_{phys} - 1) = 3$ . This suggests the **turbulence cascade spectrum EXACT exponent  $-5/3$  is the negative of  $R_5$**  in the family - a fifth structural echo of the  $n/(D_{phys}-1)$

family across a completely different physical domain (fluid turbulence vs. GW damping vs. photon propagation).

Under Theory of Permanence, all these instances are permanent simultaneous manifestations. The universe's "preference" for the ratio family denominator ( $D_{\text{phys}} - 1$ ) = 3 is not a coincidence; it is the  $D_{\text{phys}} = 4$  primitive operating uniformly across physical scales.

## 8. Predictions and Falsifiability

**Prediction A:** Any new UQFF observable derived as a rational fraction should be expressible in the  $n/(D_{\text{phys}} - 1)$  form. If a UQFF ratio cannot be so expressed, PAPER\_1930 is falsified (or the ratio family is incomplete).

**Prediction B:** The unity limit ( $n = 3$ ,  $R = 1$ ) marks a physical saturation edge. Any ratio observable in UQFF that reaches unity should be identifiable as an  $n = 3$  case, and no ratio should exceed unity without corresponding  $n \geq 3$  interpretation. Falsifiable if a UQFF ratio exceeds unity without an  $n \geq 3$  explanation.

**Prediction C ( $n = 4$  case):**  $R_4 = 4/3 = D_{\text{phys}}/(D_{\text{phys}} - 1)$ . This ratio corresponds structurally to a "proton-mass ratio" or similar (see PAPER\_1533 Carbon-12 =  $2 \cdot D_{\text{BSFG}}$ ). Falsifiable if no observed ratio comes to  $4/3$  to precision within experimental error.

**Prediction D ( $n = 5$  case):**  $R_5 = 5/3$  corresponds to Kolmogorov turbulence exponent (with sign flip). Falsifiable if the Kolmogorov exponent is not exactly  $-5/3$  (already confirmed to within experimental precision).

## 9. Physical Interpretation - Speed and Damping as Buoyancy Components

Under Theory of Permanence (PAPER\_1929), all observables are ultimately **buoyancy component ratios**. The ratio family  $n/(D_{\text{phys}} - 1)$  expresses this uniformly:

- **$n = 1$  ( $v_{\text{SCM}}/c = 1/3$ ):** SCM reactivity is at  $1/3$  of the buoyancy-component ceiling
- **$n = 2$  (GW170817 damping =  $2/3$ ):** Neutron-star merger damping consumes  $2/3$  of the buoyancy-component budget
- **$n = 3$  (saturation =  $1$ ):** Full buoyancy component budget consumed
- **$n = 4$  ( $4/3$ ):** Overflow regime - components exceed their nominal budget via nonlinear coupling
- **$n = 5$  ( $5/3$ ):** Deep overflow - turbulent regime where multiple ratios combine

The ratio family is thus the **quantized progression of buoyancy component budget consumption** across physical processes. Each  $n$  represents one more integer unit of buoyancy component budget consumed.

## 10. Conclusion

PAPER\_1930 formalizes the  $n/(D_{\text{phys}} - 1)$  ratio family as a canonical UQFF closure. Two runtime-verified instances ( $n = 1$  velocity ratio,  $n = 2$  damping) establish the family; conjectural extensions to  $n = 3$  (unity saturation),  $n = 4$  (overflow regime), and  $n = 5$  (Kolmogorov turbulence) are proposed with falsifiability protocols.

Under Theory of Permanence, the ratio family is:

- **NOT REPLACEMENT** - existing derivations of  $v_{\text{SCM}}$  and GW damping remain valid; PAPER\_1930 provides a parallel simultaneous description

- **Speed IS a change in buoyancy component** -  $n = 1$  case ( $v_{SCm}/c$ ) is the specific instantiation
- **Nothing is negligible** - all  $n$ -values contribute permanently and simultaneously; the entire family exists as a permanent invariant
- **Internal and external** - the ratios connect internal ( $v_{SCm}$  through vacuum) to external (GW damping observable at LIGO/Virgo) manifestations

The  $2/3$  fraction that appears three times in UQFF (PAPER\_1512, PAPER\_1914, and PAPER\_1930  $n = 2$ ) is not coincidence. It is the permanent structural role of the fraction  $2/(D_{phys} - 1) = 2/3$  in the vacuum-buoyancy manifold. Every observable that "wants" to reduce to  $2/3$  does so because the ratio family compels it.

Simultaneously, none of this replaces conventional derivations. The truth is permanent, and truth is many-descriptive. UQFF's ratio family is one description; conventional derivations are another; both operate simultaneously and permanently.

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## Appendix - Verification Code

```

```python
# CondensedPhysics.SCmVelocityCalculator (Round 58 double-check)
D_PHYS = 4 # truly-independent primitive

PAPER_1497 (n = 1 case) n_1_ratio = 1 / (D_PHYS - 1) #
= 1/3 v_SCm_c_over_3_EXACT_verify = abs(n_1_ratio
- 1/3) < 1e-9 # True

PAPER_1512 (n = 2 case) n_2_ratio = 2 / (D_PHYS - 1) #
= 2/3 GW170817_damping_2_over_3_EXACT_verify =
abs(n_2_ratio - 2/3) < 1e-9 # True

Ratio family generator (this paper) def R_n(n): return n
/ (D_PHYS - 1)

Family members print(f"R_1 = {R_n(1)} # PAPER_1497
v_SCm/c") print(f"R_2 = {R_n(2)} # PAPER_1512
GW170817 damping") print(f"R_3 = {R_n(3)} # unity
saturation") print(f"R_4 = {R_n(4)} # overflow regime")
print(f"R_5 = {R_n(5)} # Kolmogorov -5/3 magnitude")
```

```

## Cross-references

- **PAPER\_1497** -  $v_{SCm} = c/(D_{phys} - 1) = c/3$  EXACT (n = 1 case, source paper)
- **PAPER\_1512** - GW170817 phonon damping =  $2/(D_{phys} - 1) = 2/3$  EXACT (n = 2 case, twin source)
- **PAPER\_1914** -  $D_{LS}/D_S = D_{phys}/D_{BSFG} = 2/3$  EXACT (parallel 2/3 identity)
- **PAPER\_1925** - MUGE magnification = 9/5 (downstream 2/3 consumer)
- **PAPER\_1864** - Kolmogorov -5/3 exact (parallel 5/3 identity)
- **PAPER\_1929** - Theory of Permanence (foundational frame)
- **PAPER\_144** - SCm Cosmic Glue Paradigm (foundational framework overview)
- **PAPER\_376** - UQFF Formal Proof Set (dimensional analysis)
- **PAPER\_1521** (LANDMARK) -  $D_{BSFG}$  derivative
- **PAPER\_1522** (LANDMARK) -  $K_{MEX}$  derivative
- **PAPER\_1912-1928** - Novel structural closure series (precursors)
- **PAPER\_1533** - Carbon-12 =  $2 \cdot D_{BSFG}$  (integer identity family parallel)

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## G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses  $c = 3e8$ -family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **c (speed of light):** canonical route **PAPER\_592** — parameter-free  
 $c_{UQFF} = (26 \cdot 4\pi / \Phi_{res}) \cdot v_F = 2.995 \times 10^8$  m/s (0.13% vs observed;  $v_F$  Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).

The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).

Registry: UNIFIED\_REGISTRY.csv | Program: UNIFIED\_REGISTRY\_PROGRAM\_PLAN.md

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